Linear Graphs 4: Drawing Lines

Name _____ Date _____ No calculator Version 1

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KEY WORDS straight line *(rārangi torotika)* · gradient *(rōnaki)* · y-intercept *(pūtahi-y)*

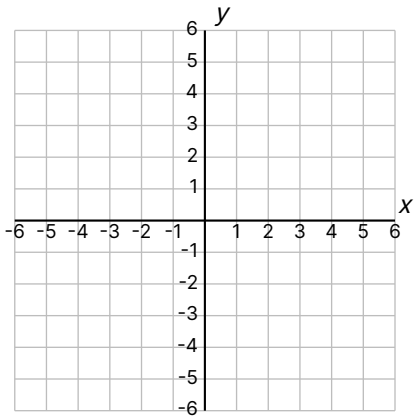
A By a table of values ●○○

Remember: Work out y for each x in the table, plot the points, then rule a straight line through them and past them to the edge of the grid.

1. Complete each table, then draw the line on the grid.

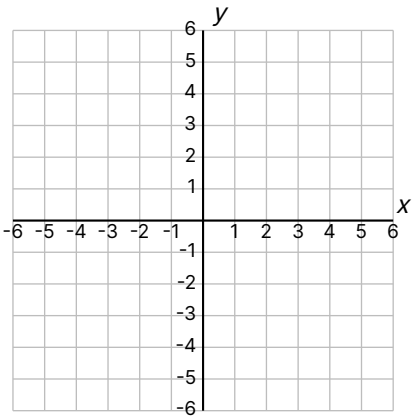
a) $y = x + 2$

x	-3	0	3
y			



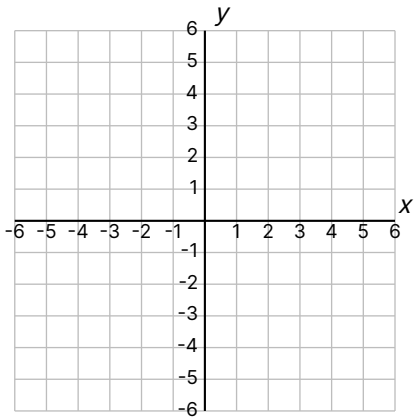
b) $y = 2x - 3$

x	-1	0	2
y			



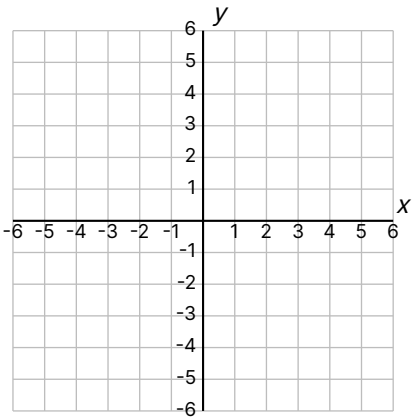
c) $y = -x + 1$

x	-2	0	4
y			



d) $y = 4 - 2x$

x	0	2	4
y			



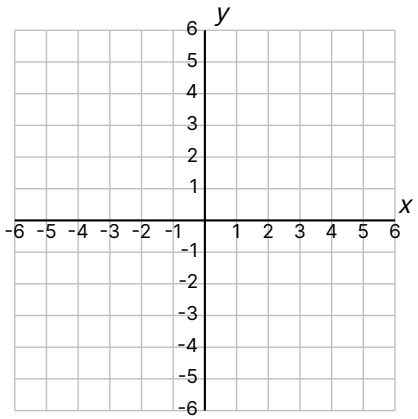
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B From the gradient and intercept ●●○

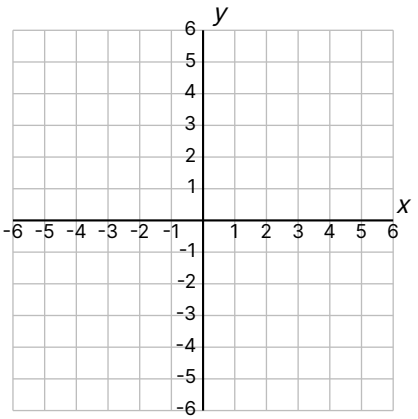
Remember: No table needed: plot the y-intercept, then step along by the gradient (for $m = 2$, go across 1 and up 2) to find more points. Rule the line and label it with its letter.

2. Draw the three lines on each grid. Label each line with its letter.

a) A: $y = 2x + 1$
B: $y = -x + 3$
C: $y = \frac{1}{2}x - 2$



b) A: $y = -3x + 2$
B: $y = x - 4$
C: $y = -2x + 3$

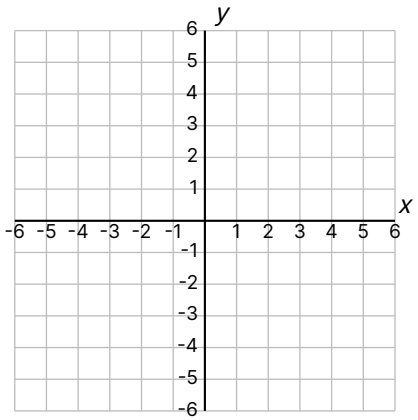


C Horizontal, vertical and through the origin ●●○

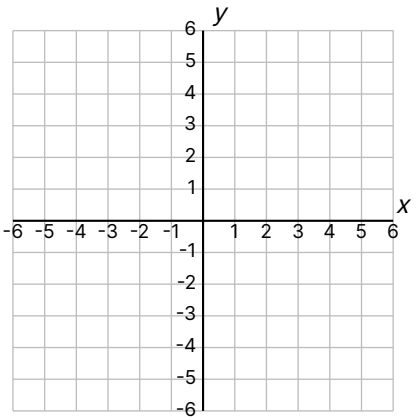
Remember: $y = 3$ is a horizontal line through 3 on the y-axis. $x = -2$ is a vertical line through -2 on the x-axis. A line with no c goes through the origin.

3. Draw the four lines on each grid. Label each line with its letter.

a) A: $y = 3$
B: $x = -2$
C: $y = 2x$
D: $y = -x$



b) A: $y = -4$
B: $x = 4$
C: $y = \frac{1}{2}x$
D: $y = -3x$



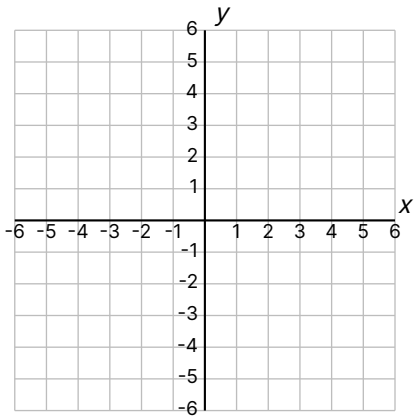
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D Fractional gradients ●●●

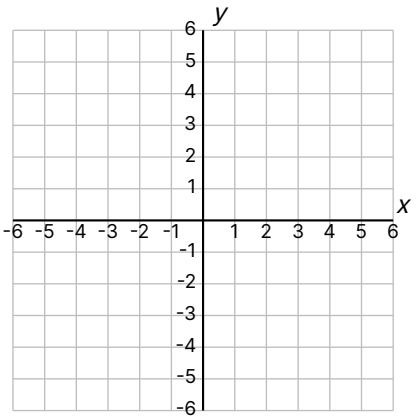
Remember: For $m = \frac{2}{3}$, go across 3 and up 2. For $m = -\frac{3}{2}$, go across 2 and down 3. Step both ways from the intercept so the line is long enough to rule accurately.

4. Draw the three lines on each grid. Label each line with its letter.

a) A: $y = \frac{2}{3}x + 1$
B: $y = -\frac{3}{2}x - 1$
C: $y = -\frac{1}{2}x + 4$



b) A: $y = \frac{3}{4}x - 3$
B: $y = -\frac{2}{3}x + 2$
C: $y = \frac{1}{3}x + 1$

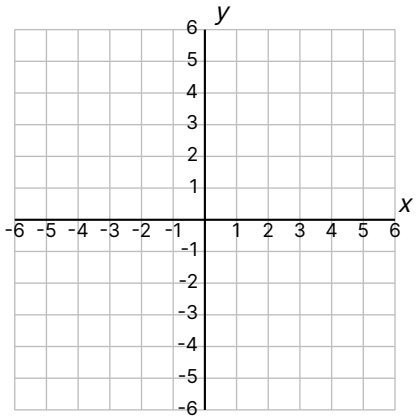


E Equations in other forms ●●●

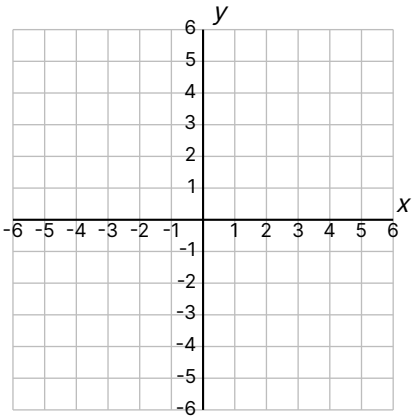
Remember: Rearrange into $y = mx + c$ first: $2y = 4x - 2$ becomes $y = 2x - 1$. For $x + y = 4$ you can instead find the two intercepts: $x = 0$ gives $y = 4$, and $y = 0$ gives $x = 4$.

5. Draw the three lines on each grid. Label each line with its letter.

a) A: $2y = 4x - 2$
B: $x + y = 4$
C: $3x - 2y = 6$



b) A: $2y = x + 4$
B: $2x + y = 5$
C: $2x + 3y = -3$

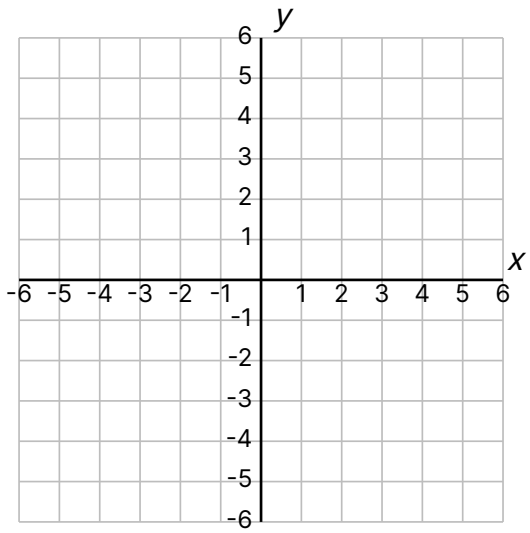


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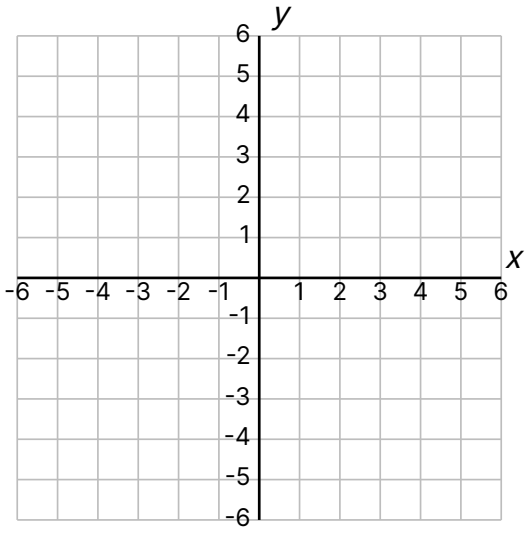
Extension

6. Draw each line on the grid, then answer in your book.

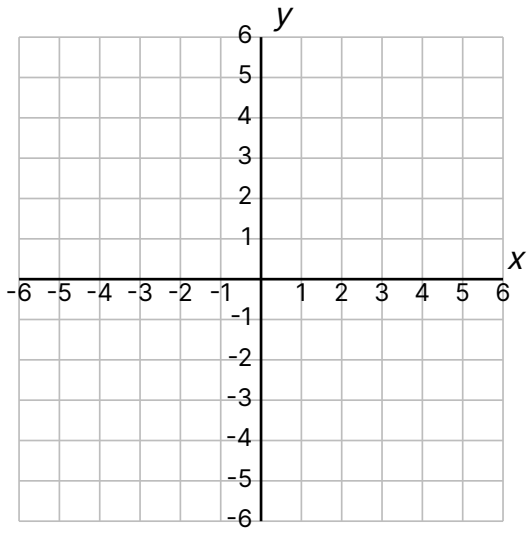
a) A: $y = x - 2$ and B: $y = -2x + 4$. Write down the point where they cross.



b) The line through $(1,\ 2)$ with gradient -2. Then write its equation.



c) The line with x-intercept 4 and y-intercept -2. Then write its equation.



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Linear Graphs 4: Drawing Lines

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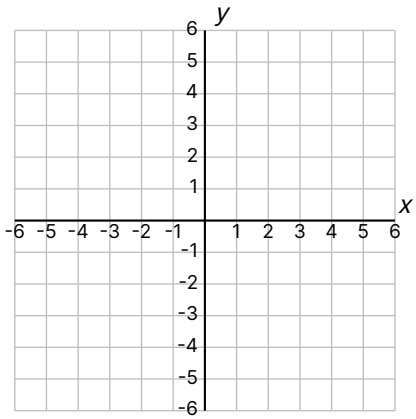
A By a table of values ●○○

Remember: Work out y for each x in the table, plot the points, then rule a straight line through them and past them to the edge of the grid.

1. Complete each table, then draw the line on the grid.

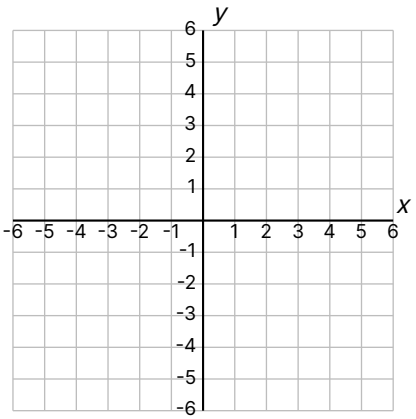
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y			



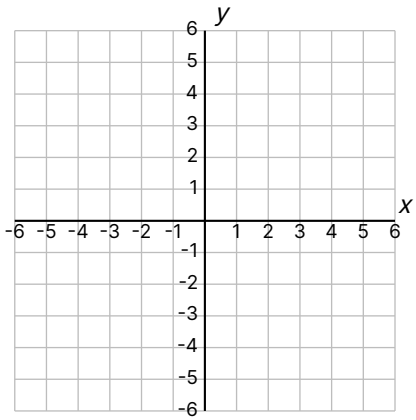
b) $y = 2x + 1$

x	-2	0	2
y			



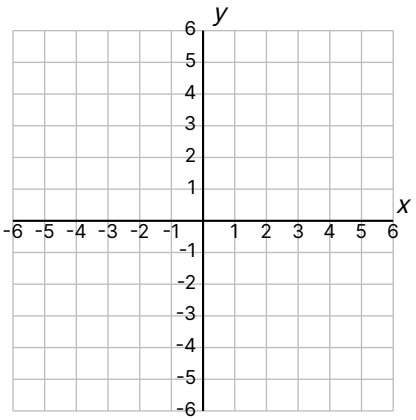
c) $y = -x + 3$

x	-2	0	4
y			



d) $y = 5 - 2x$

x	0	2	4
y			



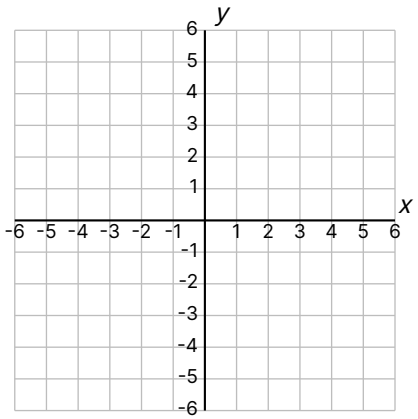
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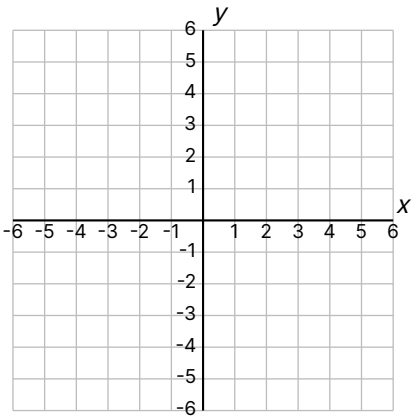
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B: $y = -x + 2$
C: $y = \frac{1}{2}x + 1$



b) A: $y = -3x + 4$
B: $y = x + 3$
C: $y = -2x - 3$

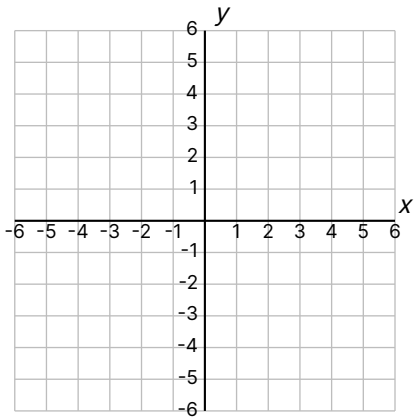


C Horizontal, vertical and through the origin ●●○

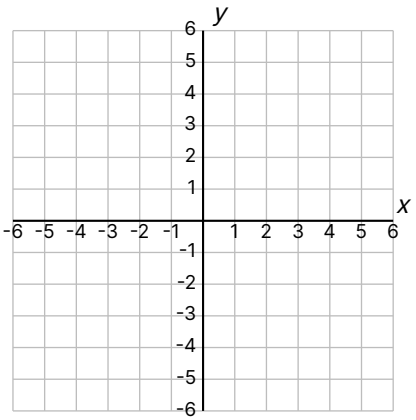
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3. Draw the four lines on each grid. Label each line with its letter.

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B: $x = 4$
C: $y = 3x$
D: $y = -x$



b) A: $y = 4$
B: $x = -4$
C: $y = -\frac{1}{2}x$
D: $y = 2x$



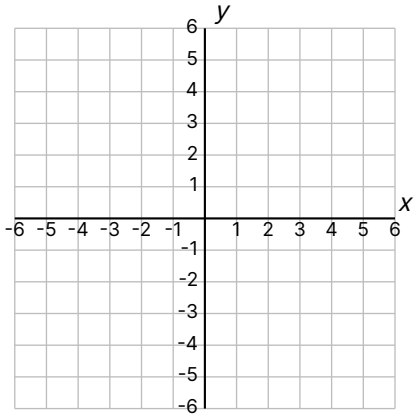
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D Fractional gradients ●●●

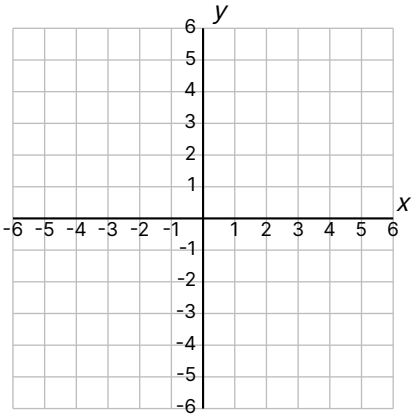
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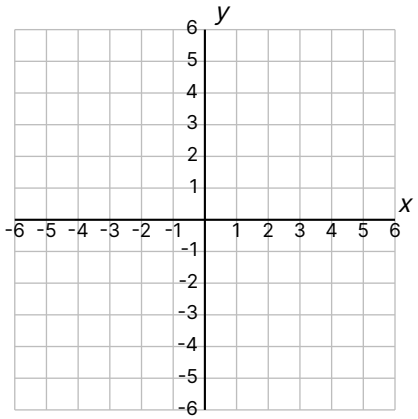


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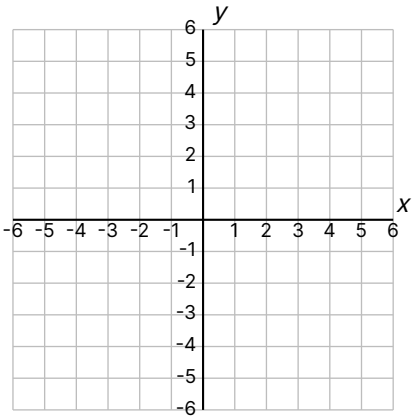
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5. Draw the three lines on each grid. Label each line with its letter.

a) A: $2y = 6x - 4$
B: $x + y = 5$
C: $2x - 3y = 6$



b) A: $2y = x - 2$
B: $3x + y = 4$
C: $3x + 4y = 12$

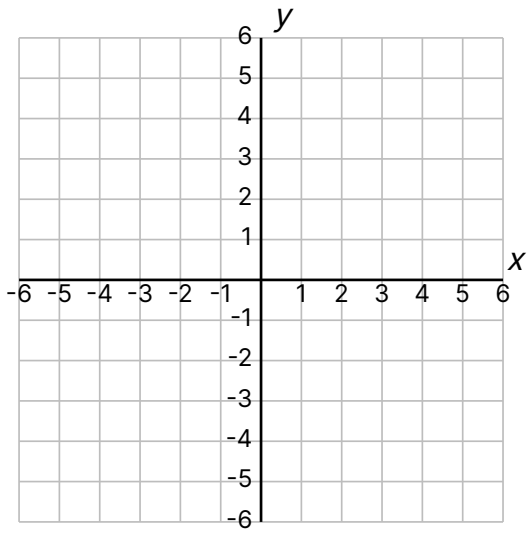


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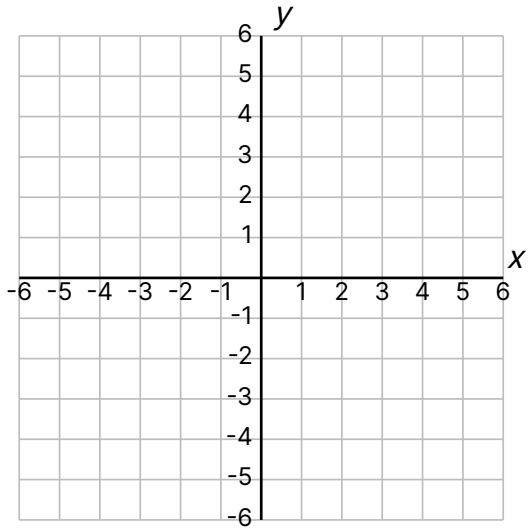
Extension

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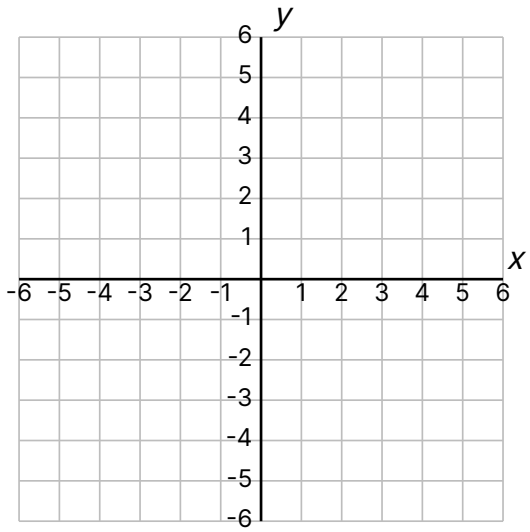
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b) The line through $(-1,\ 3)$ with gradient 2. Then write its equation.



c) The line with x-intercept -3 and y-intercept 2. Then write its equation.



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